

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285483

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

BODURKA; Alex et al.

ENTRY ACCESS SYSTEM AND METHOD WITH QUESTIONNAIRE SCREENING

Abstract

An entry system includes a plurality of first user devices, such as key fobs, ID cards and the like. A plurality of second user devices are maintained by users, and may include mobile phones, smart watches and tablets. A main entry system controls a barrier to allow entry to approved individuals. An authorization system is configured to validate the first and second user devices, and to send an electronic questionnaire to the second user device. The user must complete the questionnaire and transmit it to the main entry system. The second user device, the main entry system, and the authorization system communicate with each other over a computer network. If the user successfully completes the questionnaire and the user's identity is authenticated, the main entry system allows the barrier to be opened so that the user may enter the facility.

Inventors: BODURKA; Alex (Portage, MI), ETTRICH; Jan H. (Geneva, IL), TOLL; David (Cicero, IN)

Applicant: Masonite Corporation (Tampa, FL)

Family ID: 1000008618291

Appl. No.: 19/213562

Filed: May 20, 2025

Related U.S. Application Data

parent US continuation 18342326 20230627 PENDING child US 19213562

parent US continuation 17473492 20210913 parent-grant-document US 11727740 child US 18342326

us-provisional-application US 63077193 20200911

Publication Classification

Int. Cl.: G07C9/27 (20200101); G07C9/10 (20200101); G07C9/25 (20200101); G16H10/20 (20180101)

U.S. Cl.:

CPC G07C9/27 (20200101); G07C9/10 (20200101); G07C9/25 (20200101); G16H10/20 (20180101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND CLAIM TO PRIORITY [0001] This application is a continuation of U.S. patent application Ser. No. 18/342,326, filed Jun. 27, 2023, which is a continuation of U.S. patent application Ser. No. 17/473,492 filed Sep. 13, 2021, now U.S. Pat. No. 11,727,740, issued Aug. 15, 2023, which claims priority to U.S. Provisional Patent Application No. 63/077,193, filed Sep. 11, 2020, the disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to entry systems and methods containing identity verification and survey information, such as a wellness survey, security verification questionnaire, and the like, to control access to a buildings, secure areas or the like.

BACKGROUND OF THE INVENTION

[0003] With many contagious diseases isolation and social distancing are the most effective means to prevent the spread of the disease. For some diseases, such as COVID 19, a questionnaire may provide a rough indication of the risk of a person having the disease. For example, with COVID 19, contact with a disease positive individual, fever, respiratory symptoms, etc. may indicate that the person may have the disease. As such, office buildings sometimes question people for symptoms/risks of the disease to decide whether to allow entry. The entrants are usually required to answer the questionnaire at the door, either by a guard asking the questions or by filling out a form that is then reviewed before access is allowed. The process sometimes results in long lines at the door due to the time it takes to process the questionnaires.

[0004] Many buildings, such as office buildings or factories, already have entry systems which verify the identity of entrants before allowing entry, such as for security purposes. Such buildings would benefit from integrating a wellness questionnaire into the entry system, provided that long lines at the entrance were avoided. Therefore, there is a need for an entry system that provides identity verification as well as health screening monitoring and monitoring for regulatory, compliance and corporate governance of entrants prior to their entry into the building space.

[0005] Access to some buildings and facilities is sometimes limited for security and all regulatory requirements to individuals who have a certain credential that permit access. The credentials may be valid for a limited period, thus requiring periodic verification. Keeping track of individuals and whether they have the required credentials can be a laborious task, sometimes resulting a lines of individuals seeking access to a building or facility.

[0006] A system and method that keeps a record of individuals accessing a building or facility and also whether they have the necessary credentials in a time-saving manner is desirable in order to expedite entry and provide positive verification of who accessed the building or facility should the need arise.

SUMMARY OF THE INVENTION

[0007] The present invention provides an entry system for controlling access to a secure space, such as a building. In particular the entry system provides identity verification or authentication,

preferably multifactor identification, and a questionnaire/survey to control access to a building, secure area or the like. The questionnaire/survey may be used for health screening or to assure that individuals enter the secure space in a manner that assures regulatory compliance and/or compliance with internal corporate governance.

[0008] The entry system includes several components, such as a main entry system, a verification system, and a plurality of user devices that interact with the main entry and verification systems. The components preferably communicate with each other wirelessly to control access to the secure space.

[0009] Methods for making and using the different aspects of the present invention are also provided.

[0010] Other aspects of the invention, including apparatus, devices, kits, processes, and the like which constitute part of the invention, will become more apparent upon reading the following detailed description of the exemplary embodiments

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The following detailed description of the invention will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there is shown in the drawings embodiments which are presently preferred. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:

[0012] FIG. 1 is a schematic of the entry system of the present invention;

[0013] FIG. 2 is a perspective view of an exemplary entranceway;

[0014] FIG. 3 is a diagram illustrating the entranceway of FIG. 2 with geofencing technology;

[0015] FIG. 4 is a flow diagram showing the system as configured for use during an epidemic to control access by a user (employee) at a main entrance;

[0016] FIG. 5 is a continuation of the flow diagram of FIG. 4;

[0017] FIG. 6 is a flow diagram showing the system configured for use (to monitor general wellness or a specific disease or condition) to control access by a user (visitor) at a main entrance;

[0018] FIG. 7 is a continuation of the flow diagram of FIG. 6;

[0019] FIG. 8 is a diagram showing the system configured for use to monitor a user's (employee's) general wellness and to control access by the user at a main entrance;

[0020] FIG. 9 is a diagram showing the system configured for use to monitor a user's (employee's) general wellness and to control access by the user at a side entrance;

[0021] FIG. 10 is a diagram showing an exemplary entry system;

[0022] FIG. 11 is a diagram showing the data flow in the exemplary entry system for questionnaire delivery;

[0023] FIG. 12 is a flow chart showing operation of the exemplary entry system;

[0024] FIG. 13 is a flow chart showing the questionnaire configuration process;

[0025] FIG. 14 is a flow chart showing the authorization configuration process;

[0026] FIG. 15 is a flow chart showing a method for delivery of the questionnaire using geofencing;

[0027] FIGS. 16a and 16b are flow charts showing a method for delivery of the questionnaire using passive pointers (FIG. 16b is a continuation of FIG. 16a);

[0028] FIG. 17 is a flow chart showing a method for delivery of the questionnaire using broadcast pointers;

[0029] FIGS. 18a and 18b are a flow charts showing a method for delivery of the questionnaire using text messaging (FIG. 18b is a continuation of FIG. 18a);

[0030] FIG. **19** is a flow chart showing the identity input configuration process the user must follow to gain access to the secure space
[0031] FIG. **20** is a flow chart showing the authentication input using the second user device;
[0032] FIG. **21** is a flow chart showing the authentication input using alphanumeric value;
[0033] FIG. **22** is a flow chart showing the authentication input using proximity of the second user device to the main entry system;
[0034] FIG. **23** is a flow chart showing the authentication input using biometric data of the user;
[0035] FIG. **24** is a flow chart is a flow chart showing the authentication input using biometric authentication input from the user device;
[0036] FIG. **25** is a flow chart showing the authentication input using biometric authentication input from the authentication input device **1312** of the main entry system
[0037] FIG. **26** is a flow chart showing the authorization check process.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0038] Reference will now be made in detail to exemplary embodiments and methods of the invention as illustrated in the accompanying drawings, in which like reference characters designate like or corresponding parts throughout the drawings. It should be noted, however, that the invention in its broader aspects is not limited to the specific details, representative devices and methods, and illustrative examples shown and described in connection with the exemplary embodiments and methods.

[0039] The present invention relates to entry systems for allowing entry into a secure space, such as an office building, a factory, a manufacturing facility, etc. Referring to FIG. **1**, the entry system **100**, preferably an automated system provided with a hardware and software platform, includes a main system **102** in electronic communication with an identity verification system **104** for verifying the identity of the building entrants, and interfacing with a plurality of user devices **106**, such as fob keys, smartphones, tablets, watches and the like.

[0040] Part of the identity verification system **104**, preferably includes a verification device **105**, as shown in FIG. **2**, that may be located at the entrance to the secure space, such as the entrance to a building, laboratory, etc. The entrance may include a frame **108** mounted to a wall around the entrance, and a barrier, such as a door, **110** pivotally or movably mounted on the frame **108**. The verification device **105** may be mounted to the frame or the barrier **110**. The barrier **110** can partially or completely cover the frame opening (FIG. **2** shows the barrier **110** partially covering the frame opening) and can be, but is not limited to, a door, a gate, a bar, a curtain, a turnstile, or combinations thereof. Other visual or physical restraints may also be appropriate for the present invention.

[0041] The identity verification device **105** may be, but is not limited to, an RFID reader, a fingerprint scanner, a retinal scanner, a camera, a keypad, a writing pad, a voice recorder, a bar code reader, a QR reader or combinations thereof. Identity (ID) verification may, e.g., include the capability of capturing images or data from or about individuals, and of comparing the image or data to data within existing internal databases or third-party databases for identity matches. ID verification may be capable of capturing images of a user/person, or reading his/her identification card, employee badge, a bar code, a QR code, etc., and then cross referencing the images with the database of a third-party access control system (ACS). A photo of a person may be analyzed by a facial recognition algorithm to compare it against a database of images. Upon a match in the data or a positive identification of the individual person, the system may allow the person to enter the secure space.

[0042] In another embodiment, multi-factor verification may be used to improve the reliability of ID verification. Multi-factor verification or multi-factor authentication may be accomplished by requiring a user's entry privileges to be authenticated by using more than just a simple key card. At many secure facilities, users are provided with access badges to swipe against door readers to enter restricted areas. One problem with such access control is that anyone that has access to the badge

can improperly gain access to the restricted area. Multi-factor verification minimizes this problem by validating the identity of the user that is requesting access twice, e.g., once with a key card or the like, and then a second time with facial recognition, fingerprints, iris scan, face/voice recognition, signature, or personal identification number.

[0043] A user device **106**, such as a smart phone, a smart watch, or a tablet is to be associated with the user. The user device **106** is provided with a hardware and software platform for electronic communication with the main system **102**. The user device **106** preferably communicates wirelessly with the main system **102**. The user device **106** communicates with the user interface electronically through electronic communication devices that may include sensors, gateways, switches, routers, hubs or any other electronic communication device to the remotely located main system **102**. The user device **106** may transmit its information, e.g., location and identity of the user, on an ongoing basis or intermittently (preferably intermittently), directly to the main system **102** or via the other electronic communication devices. Preferably, the user device **106** communicates with the main system **102** only when the user device is within specified areas (as discussed below). The communication preferably uses wireless technologies and protocols for electronic communication such as cellular (3G/4G/5G), Bluetooth, Bluetooth Low Energy, WiFi, TCP/IP, near field communication (NFC), and other such technologies and protocols.

[0044] In a preferred embodiment, geofencing may be used to trigger a push notification to the user device **106** advising the user to fill out a questionnaire before attempting to enter the secure space. The user device **106** communicates its location information to the main system **102** on some basis, such as periodic or intermittent. Referring to FIG. 3, when the transmitted location is within a specified location or area **112** relative to the secure location **114**, the main system **102** is notified. The specified location **112** may be set in advance in the main system **102**, and may be, e.g., the parking lot of the secure space **114** and/or a specified distance from the secure space **114**. For example, as shown in FIG. 3, the specified location may be an area within a radius R from the center of the secure space **114**. When a user is within the specified location **112**, as determined by the location of the user device **106**, the main system **102** sends a push notification to the user to fill out the questionnaire. The user may be instructed to log in to a website to answer the questionnaire or to answer the questionnaire sent directly to the user device. Once the questionnaire is completed, it, along with the user identity information, is forwarded to the main system **102**.

[0045] In case of COVID-19, the questionnaire includes questions relating to risk factors indicating high risk of contracting the disease or symptoms of the disease. Of the symptoms, the questions may ask whether the user has one or more of the following symptoms: [0046] Fever or chills [0047] Cough [0048] Shortness of breath or difficulty breathing [0049] Fatigue [0050] Muscle or body aches [0051] Headache [0052] New loss of taste or smell [0053] Sore throat [0054] Congestion or runny nose [0055] Nausea or vomiting [0056] Diarrhea [0057] Loss of smell or taste [0058] Other the risk factors include whether the user has [0059] Been tested for the disease [0060] Been diagnosed with disease [0061] Been in contact with a person with disease within the last 14 days. [0062] Travelled to any regions affected by disease within the last 14 days Underlying health conditions, such as heart disease, lung disease, kidney disease, diabetes, autoimmune disorders, etc.

[0063] Other diseases, including other infectious diseases, mental health, or overall wellness, may also be similarly screened using similar questionnaires with risk factors and symptoms associated with that disease. The questionnaire may inquire about the user's mental state (angry/sad/happy), stress, or other conditions. For infectious diseases, the user should access the questionnaire before requesting entry to the secure space **114**. For general wellness and mental health, the user should access the questionnaire periodically, once a week or once a quarter.

[0064] While we disclose use of the system **100** for health screening, it may be used to assure that individuals enter in a manner that assures regulatory compliance and compliance with internal corporate governance. The system **100** utilizes observed data (biometrics) and user-provided data to

determine whether to perform some action, such as allow access, track wellbeing, provide a notification, etc. Security regulations may, for example, preclude entrance where an individual has visited a country or region within a period of time. Corporate governance may preclude entrance until an entrant has signed a non-disclosure agreement (NDA). The system **100** allows multiple types of questionnaires to be provided to a proposed entrant in order to assure compliance with health, regulatory and corporate governance protocols. Regulatory and corporate governance may include updating or providing appropriate non-disclosure agreements (NDAs), security clearance, safety protocols, codes of conduct, non-compete, engagement census, and other re-accruing processes that require acknowledgment or feedback from employees or visitors.

[0065] In addition to the questionnaire, the system **100** may include one or more sensors **116** (FIG. 2), such as a temperature sensor, a camera, other the like. For example, the sensor **116** may be a temperature sensor to detect body temperature of the user. Alternatively, the sensor **116** may be a camera with associated software to detect the user's facial features to determine the user mood (angry/sad/happy).

[0066] Once transmitted to the main system **102**, the questionnaire is valid only for a selected amount of time. For example, the selected amount of time may be 10 minutes or less, 5 minutes or less, preferably about 2 minutes. The selected amount of time is preferably calculated to allow the user to arrive at the entrance of the secure area to request entry. While health screening may require that the period during which a questionnaire response is valid is relatively short, other protocols, such as for regulatory and corporate governance may allow a much longer period, such as a calendar quarter or possibly longer. After the selected amount of time has passed, the answered questionnaire expires. If the user is still in or near the specified location **112**, the main system **102** notifies the user device **106** of the expiration and requests that another questionnaire be completed. Alternatively, the main system **102** may notify the user device **106** that the questionnaire is expiring and request that the user affirmatively extend the expiration.

[0067] Operation of the entry system **100** is described in the following. When the user is within the specified location **112**, the user device **106** notifies the main system **102**, e.g., via an application installed on the user device **106** that is in communication with the main system **102**. The user is then advised to fill out the questionnaire. Alternatively, the user may fill out the questionnaire before leaving for the secure space **114**. One advantage of filling out the questionnaire before leaving is that the user is informed whether he/she is cleared for entry before having to drive all the way to the secure space **114**. Once completed, the questionnaire and user information are transmitted to the main system **102**. Preferably, if the questionnaire is not cleared, the user is immediately notified that entry will not be permitted. When the user first requests entry into the secure space **114** and presents his/her credential, such as a badge containing an RFID chip, the identify verification device **104** reads the badge or other credential and confirms the identity of the user. At the same time, the main system **102** confirms whether the user has filled out the questionnaire. The person's authorization to enter is validated only when the person's identity and questionnaire are cleared. Once validated, the person is granted access and entry to the secure space **114**. To be granted access to the secure space **114**, the person's identity must be verified, and the questionnaire must be cleared, i.e., no risk or symptom of COVID-19 (or other infectious disease). If one or more of the symptoms or one or more risk factor are present, entry may be denied. If entry is denied, in certain embodiments, a receptionist/attendant/healthcare professional is called to perform further testing, perform further questioning, and/or to reschedule entry for a later time. The receptionist/attendant/healthcare professional may meet the user at the entrance to the secure space **114** or contact the user at a later time.

[0068] When epidemic disease condition is not prevalent, the system **100** may be used to monitor the user general well-being. For example, the system **100** may be used to detect the individual's general wellness. If unwell conditions are detected by the system **100**, reminders may be sent to the user to take a day off and/or to seek healthcare.

[0069] FIGS. **4-5** illustrate use of the system **100** when an employee wishes to access the main entrance of the office or like work facility. In that event, the employee/user may take the survey from home and access to the facility will be permitted when the survey is successful and a key card or like identifier swiped at the entrance determines the identity of the employee and that the employee has successfully passed the survey. If the survey has failed, then access is denied.

[0070] FIGS. **6-7** illustrate use of the system **100** when a visitor seeks to access the facility. In that event, the visitor's identity is first determined, such as by presentation of a valid driver's license or other government identification card, and then the survey completed successfully prior to access being allowed.

[0071] FIGS. **8-9** illustrate use of the system **100** when being used to monitor the general wellness of employees. The operation is similar to FIGS. **4-5**, except that the survey questions relate to the general well-being of the individual and are not directed to a particular disease.

[0072] FIG. **10** shows an exemplary entry system **1000** of the present invention utilizing multifactor verification. The system **1000** includes a network **1100** operating over cellular, WiFi, and/or wired Ethernet networks and GPS system. The system **1000** includes first user devices **1006a**, second user devices **1006b**, a main entry system **1002**, a verification (or authorization) system **1004**, and an administrator device **1104**. FIG. **11** shows the communication and data flow of the system **1000**.

[0073] The network **1100** preferably uses wireless technologies and protocols for electronic communication such as cellular (3G/4G/5G), Bluetooth, Bluetooth Low Energy, WiFi, TCP/IP, near field communication (NFC), and other such technologies and protocols. Preferably, the network **1100** also provided with location information from the second user device **1006b** via the GPS on the second user device **1006b**. The components of the system communicate with each other via the network **1100**.

[0074] The system **1000** preferably uses two user devices **1006a** and **1006b** for two-factor authentication. The first user device **1006a** may preferably be, but not limited to, a FOB key, a card key, or other devices using RFID technology, for example, to enter a building. The first user device **1006a** need not interface with the network **1100**, as it may be read by the verification system **1004** at the entrance to the secure space **114** or on the pathway to the secured space **114**. Preferably, the first user device **1006a** provides a unique identifier for every user, so that the main entry system **1002**, on reading the first user device **1006a**, identifies the first user device **1006a** as associating with a particular user.

[0075] The second user device **1006b** may preferably be a smart phone, a smart watch, laptop, or a tablet associated with the user. The second user device **1006b** preferably includes one or more of the following modules: a network interface **1200**, a user interface **1202**, an operating system (OS) notification handler **1204**, a location service **1206**, a camera **1208**, a microphone **1210**, a browser **1212**, a cellular messenger **1214**, and a questionnaire viewer application **1216**. A typical smart phone typically includes all of those modules.

[0076] The network interface **1200** interfaces the second user device **1006b** to the network **1100** wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks, and is usually provided by the operating system of the second user device **1006b**. The network interface **1200** is used by the device **1006b** to communicate with the main entry system **1002** and the verification system **1004** via the network **1100**.

[0077] The user interface **1202** is preferably provided by the OS of the second user device **1006b** and provides a platform for the user to input information to applications running on the second user device **1006b**.

[0078] The notification handler **1204** is provided by the OS of the second user device **1006b** and handles the receiving and presentation of push notifications to the device **1006b** (see FIG. **11**). The notification handler **1204** is used by the questionnaire viewer **1216** to let the user know he/she has a questionnaire to fill out (see FIG. **11**).

[0079] The Location Service **1206** provides location information to applications running on the second user device **1006b**. It is preferably provided by the second user device **1006b** and may use GPS, WiFi, Bluetooth, and other triangulation methods for providing the location of the second user device **1006b**. The questionnaire viewer **1216** preferably uses this location service **1206** in two ways: 1) to automatically present the user with a questionnaire based on the location or proximity to an entry system; and 2) to use the location service **1206** as an authentication input as it provides additional assurance that a particular user is the one who is attempting to gain access (multifactor authentication).

[0080] The camera **1208** is preferably provided by the second user device **1006b** and operable via the device's OS. The user may use the camera **1208** to scan a QR Code to launch a questionnaire. Additionally, the questionnaire viewer **1216** can also use the camera **1208** to provide an authentication input in the form of an image of the user's face, e.g., for facial recognition for use in multifactor authentication (see FIG. **11**).

[0081] The microphone **1210** is preferably provided by the second user device **1006b** and operable on the device's OS. The questionnaire viewer **1216** can also use this to provide an authentication input in the form of a recording of the user's voice, e.g., for voice recognition for use in multifactor authentication.

[0082] The browser **1212** is preferably provided by the second user device **1006b** and operable on the device's OS. This browser can be used by the user to fill out the questionnaire.

[0083] The cellular messenger **1214** may be a text message (a.k.a. SMS) service provided by the second user device **1006b** and operable on the device's OS. A user may use this cellular messenger **1214** to request a URL to a questionnaire from the system **1000**.

[0084] The questionnaire viewer **1216** is preferably an application on the second user device **1006b** that the user uses for filling out and managing questionnaires. It can also configure the OS to provide notifications and location tracking used by the questionnaire viewer **1216** to automatically present users with the questionnaire.

[0085] The main entry system **1002** allows entry into the secure space **114**. It preferably includes processors and associated electronics that are purpose-built for the application, such as the access controller, questionnaire pointer, notification handler as explained below; however, off-the-shelf electronic components, such as the barrier, authentication input device, identity input device, network interface, and database as explained below, may be included. The main entry system **1002** preferably includes one or more of the following modules: a network interface **1300**, a notification handler **1302**, a database **1304**, a barrier **1110**, an access controller **1306**, a hash generator **1308**, an identity input device **1310**, and an authentication input device **1312**.

[0086] The network interface **1300** interfaces the main entry system **1002** to the network **1100** wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks. The network interface **1300** allows the main entry system **1002** to communicate with the network **1100**, and thereby, with the second user device **1200** and the verification system **1004**.

[0087] The notification handler **1302** allows the main entry system **1002** to provide user feedback (i.e., notifications). These notifications may provide reminders to the user of actions need by the user. For example, the notifications may let the user know that a questionnaire will expire in the near future or has expired but is not needed for access. Additionally, for health screening, the notifications may indicate to the user whether he/she has a health metric that is out of the normal range. Other notification actions may be providing an NDA, additional security clearance, health screening, safety protocols, codes of conduct, non-compete, engagement census, or other re-accruing processes that require acknowledgment or feedback from employees or visitors.

[0088] The database **1304** is a storage device used by the access controller **1306** to store system configuration data, history logs, user data objects, and other data required for proper operation.

[0089] The barrier **1110** provides a physical barrier used to grant and restrict access to the secure space **114**. A typical barrier **1110** is a door having a latch for locking and unlocking of the door. The

access controller **1306** electronically controls the locking and unlocking of the latch to allow or restrict access to the secure space **114** via the barrier **1110**.

[0090] The access controller **1306** is preferably an electronic device equipped with a processor, memory, storage, and software application(s). The access controller **1306** receives local inputs from the authentication input device **1312** and the identity input device **1310**. In addition, the access controller **1306** communicates with other devices on the network via the network interface **1300**. The access controller **1306** is responsible for controlling the barrier **1110** which controls access to a physical space. Furthermore, the access controller **1306** may be used to control the QP **1004** and notification handler **1302**.

[0091] The hash generator **1308** (a.k.a. hashing function, cryptographic hashing algorithm) is used to convert and securely store sensitive biometric (e.g., facial imagery and voice) data. The hash generator converts data of an arbitrary size to data of a fixed size. This compresses the data to significantly reduce the amount of storage required to store the data. It also reduces the amount of data that has to be shared over the network **1100**. A cryptographic hash algorithm for the hash generator provides additional security and privacy benefits since the user's raw biometric data is never stored. The hash generator **1308** may be a separate module (as shown in FIG. **10**) or on the same electronic board as the access controller **1306**.

[0092] The identity input device **1310** is used to capture the identity input from the user. A preferred implementation involves an RFID scanner for reading the first user device **1006a** or a keypad for entry of an entry code (e.g., a combination of letters and/or numbers uniquely assigned to the user) from the user. This input is then sent to the access controller **1306**.

[0093] The authentication input device **1312** captures an authentication input from a user. The authentication input device **1312** may include a digital camera and/or a microphone that can be enabled/disabled by the system administrator via the configuration of the system, e.g., for multi-factor authentication. The authentication input device **1312** captures raw imagery of a user's face and/or voice, which are sent to the access controller **1306** and eventually to the authorization system **1004** to authenticate a user through facial or voice recognition as explained below.

[0094] The questionnaire pointer (QP) **1314** is a sign or display, preferably controlled by the access controller **1306** and provides a link to a questionnaire required for accessing the building. The QP **1314** preferably provides a URL link to a questionnaire/survey/form that the user must fill out before entry into the secure space **114**. The QP is preferably an electronic sign that provides the URL link by providing a QR Code or phone number for receiving the URL via text message. Additionally, the QP **1314** can be used for authentication purpose. The QP **1314** preferably includes three modules: a digital display, a static code (e.g., a non-digital sign), and a radio. The digital display allows the main entry system **1002** to display text and imagery that are electronically sent from the access controller **1306**. When used for delivering a questionnaire to a user, the display can be configured to display a QR Code, URL, or phone number (see FIG. **11**). These allow the user to access the questionnaire/survey/form that the user is required to complete, preferably via the second user device **1006b**. When used for authentication purposes, the digital display can provide a one-time-use alphanumeric value to the user. The user would then use the questionnaire viewer **1216** on the second user device **1006b** to enter this value. Providing a correct alphanumeric value authenticates that a particular user is attempting to use a specific entry system at a particular point in time.

[0095] The static code is used to provide a URL link to a questionnaire/survey that visitors of a building must fill out before entry. The QP **1314** can provide the URL link by giving a static QR Code or phone number for receiving the URL via text sent to the second user device **1006b**. The radio is electronically controlled by the access controller **1306**. When used for delivering a questionnaire to a user, the radio can broadcast a URL that can be scanned by a user's phone. This can be done via Bluetooth, Bluetooth Beacons, or NFC. Additionally, the radio may be a WiFi radio, which is configured as an access point with no password required. However, when the user

attempts to join the network, the user is presented with a questionnaire instead of a login screen. Once the user submits the questionnaire, the user's device will be removed from the network (the user cannot use this network to connect to the internet). When used for authentication purposes (multifactor authentication), the radio can be used to determine whether the second user device **1006b** is within close proximity, preferably less than about 100 m, more preferably less than about 50 m, most preferably less than about 20 m to the barrier **1110** where the identity code is entered by the user. In this scenario, the radio broadcasts itself to be scanned by the second user device **1006b** via the questionnaire viewer **1216**. The questionnaire viewer **1216** then sends this information (whether the second user device **1006b** is proximate to the barrier **1110**) to the authorization system **1004**. If the second user device is indeed proximate to the barrier **1110**, the authorization system **1004** authenticates the user who entered the identity input information into the main entry system **1002**.

[0096] The verification system (authorization system) **1004** preferably includes one or more servers that can be installed on-premise or remotely. For latency concerns, local installation is preferred. However, a remote server may provide a less expensive alternative. The authorization system **1004** is used for hosting, managing, and configuring a questionnaire application **1410** located thereon. Additionally, the system determines whether a particular user is authorized to enter a building or area within a building. The authorization system **1004** verifies the identity and authentication inputs that the user presented to the main entry system **1002** and/or the second user device **1006b**. Furthermore, the authorization system **1004** may require the user to submit a questionnaire before authorizing access by the user. The authorization system **1004** takes the user's inputs and determines whether or not the user has presented the correct information to be authorized for access. The authorization system **1004** preferably includes one or more of the following modules: a network interface **1400**, a database **1402**, a messaging service **1404**, an identity and authentication application (IAA) **1406**, a hash generator **1408**, the questionnaire application **1410**, and an authorization application **1412**.

[0097] The network interface **1400** interfaces the authorization system **1004** to the network **1100** wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks. This network interface **1300** allows the authorization system **1004** to communicate with the network **1100**, and thereby, with the second user device **1006b** and the access controller **1306** on the main entry system **1002**.

[0098] The database **1402** is a storage device used by the authorization system **1004** to store system configuration data, history logs, user data objects, questionnaires, questionnaire submissions, and other data required for proper operation.

[0099] The messaging service **1404** allows the user to send a text message to request a link to the required questionnaire. The user may be notified to access the questionnaire, e.g., by a notification push to the second user device **1006b** or by instruction from the authorization system **1004** or the main entry system **1002**. The user is instructed, e.g., by the QP, to provide his/her location by indicating the position ID in the body of the text. The messaging service **1404** parses the user's text for the location identification (ID). With the location ID the messaging service will then send this to the questionnaire application (see below) to determine which QP is associated with that location ID and obtain a URL to those questionnaires. The messaging service **1404** then sends the questionnaire(s) link(s) back to the user, preferably via SMS, with link(s) to the questionnaire(s) associated with that location (see FIG. 11).

[0100] The identity and authentication application (IAA) **1406** is configured to manage new identity and authentication inputs, and to ensure that new identity inputs (manual codes or unique identifiers of the first user device) are unique and never re-used. Additionally, The IAA **1406** also handles the secure storage of biometric data used for authentication purposes (see FIG. 11).

[0101] The hash generator **1408** (a.k.a. hashing function, cryptographic hashing algorithm) is configured to convert and securely store sensitive biometric data. The hash generator takes data of arbitrary size to data of a fixed size. This compresses the data to significantly reduce the amount of

storage required to store data. It also reduces the amount of data that has to be shared over a network. A cryptographic hash algorithm for the hash generator provides additional security and privacy benefits, as the user's raw biometric data is never stored on the system. Although FIG. 10 shows the hash generator **1408** as a separate module, it may also be part of the IAA **1406**.

[0102] The questionnaire application **1410** is configured for the creation, management, and hosting of questionnaires. Administrators can remotely access the application **1410** for managing the questionnaires and export submissions for data-keeping purposes. The user can access the questionnaires hosted by the questionnaire application **1410** via a URL or directly in the questionnaire viewer running on his/her second user device **1006b**. The questionnaire as described above and provides a list of questions used to aid in health screening, regulatory, compliance, and/or corporate governance. The questionnaire is preferably saved on the database **1402**. The questions from the questionnaire are requested to be filled out by the user. Each or some of the questions may be configured to be used as an additional authorization check for determining authorization. The user's response for question(s) used for authorization check must match a list of acceptable answers (provided by the administrator at time of questionnaire creation) to receive authorization to enter the secure space **114**. Otherwise, authorization is denied. The questionnaire may be delivered to the user in different ways. The system **1000** may disable/enable functionalities based on the deliver method selection based on the system configuration for the particular user. The delivery of the questionnaire may be through geofencing as discussed above. When the second user device **1006b** is within a pre-defined perimeter, preferably within about 50 m the secure space **114**, any required questionnaire(s) is automatically sent to the second user device **1006b**. The location is determined by the OS location services running on the second user device **1006b**. The location may also be determined by WiFi, Bluetooth, GPS, or other triangulation methods. The delivery of the questionnaire may also be through the QP **1314** or the remote questionnaire pointer (RQP) **1102** (as describe below). The QP **1314** or RQP **1102** are devices that provide a questionnaire URL to the user. Depending on the capabilities of the device that is providing the URL, there are two types: passive QP and broadcast QP. Passive QPs require the user to manually perform an action to get a link to the URL. For example, the user can scan a QR code, request a URL via SMS, or type in a URL displayed by the device. Broadcast QPs use devices equipped with radios which broadcast the URL. The questionnaire viewer **1216** running on the second user device **1006b** receives the radio broadcast when it is within radio range, and notifies the user that a questionnaire must be completed to access the secure space **114**. The radio broadcast may be implemented with WiFi, Bluetooth, or NFC. The user's responses expire on after a predetermined amount of time or when the questions are no longer in use, whichever comes first. The predetermined amount of time depends on the particular question and may be from about 1 day to about several years. For example, for a COVID 19 questionnaire, the expiration may be daily. On the other hand, for an NDA, the expiration may be determined by the language of the NDA. A list of devices used as QPs (RQPs **1102** and QPs **1314**) associated with a particular questionnaire are also saved in the database **1402**. The URL provided to the user preferably contains an identifier to allow the authorization system **1004** to know which device is used to obtain the URL. If the device is not associated with a questionnaire, as set by the system configuration, the user is not presented with a questionnaire. A list of users who have completed the questionnaire is also saved on the database **1402**.

[0103] The authorization application **1412** is used to configure the authorization checks required for a particular user or user group. The configuration may be dome by the system administrators. The authorization application **1412** creates and maintains the user data object saved on the database **1402**. The authorization application **1412** uses the user data object to grant access to the user. When the user requests access, the main entry system **1002** only grants access if the authorization application **1412** determines all of the user's authorization checks have been met. The user data object is unique to each particular user and used by the authorization system **1004** to determine authorization access, and inform the system how the user can provide authentication/identity inputs

to the system. The particular information contained in the user data object is explained below. The authorization system **1004** also maintains a list of the authorization checks required for each particular user to gain access to the secure space **114**. For all users, however, identification is always required to gain access. Two other authorization checks may also be required of the user to gain access: an authentication input (to authenticate the user entering the identity input) and/or questionnaire responses. The additional authentication may be facial or voice recognition, for example. If additional authentication is required, the hashed value of the raw authentication input (facial or vocal), if that input method is selected, is generated and saved in the database **1402**. The authorization application **1412** may also be used to define input methods for each particular user. Depending on the particular user, the authentication and identity input methods may differ. For example, a CEO may be required to have a more secure method of authenticating themselves when compared to a janitor. Depending on the user, no authentication may be necessary or one or more input methods may be required (depending on the level of security required to enter the secure space **114**).

[0104] One authentication input method is approval from the second user device. Here, when the user uses his first user device **1006a**, e.g., at the barrier **1110** or his assigned code (as described below), the user is presented with a notification that their identity is being used at the particular barrier **1110**. This notification is delivered via the questionnaire viewer **1216** running on the second user device **1006b**. The user must then approve that they are the one using the first user device **1006**. If the user does not approve this within a specific time window, for example within **60** seconds, the process must be restarted.

[0105] Another authentication input method uses an alphanumeric value comparison method. Here, when the user uses the first user device **1006a**, e.g., at the barrier **1110** or the assigned code (as described below), the main entry system **1002** displays an alphanumeric value to the user, e.g., on the digital display of the QP **1314** located proximate to the barrier **1110**. The user must enter this value into the questionnaire viewer **1216**, which is then sent to the authorization application **1412** for comparison. If the user-provided input matches the value displayed to him/her, this authorization check is complete.

[0106] A further authentication input method uses proximity of the second user device to the radio module on the main entry system **1002**. Here, when the user uses the first user device **1006a** or the assigned code (as described below), e.g., at the barrier **1110**, the radio module on the QP proximate to the barrier **1110** is configured to broadcast. If the second user device **1006b** is within proximity of the radio, the questionnaire viewer **1216** running on the second user device **1006b** communicates to the authorization system **1004** that this user is in close proximity to the entry system **1002**, which may be used as an authentication input.

[0107] Yet another input method uses biometric data, such as facial recognition and/or voice recognition. Here, when the user uses the first user device **1006a** or the assigned code (as described below), e.g., at the barrier **1110**, the main entry system **1002** requests a facial and/or voice recording from the user, preferably via the questionnaire viewer **1216** running on the second user device **1006b**. The questionnaire viewer **1216** may capture the user's facial image by the camera **1208** on the second user device **1006b**. Alternatively, the authentication input on the main entry system **1002** can be used for image capture. The camera may be mounted near the barrier **1110**. For voice recognition, the user's voice may be recorded by the questionnaire viewer **1216** using the microphone **1210** on the second user device **1006b**. Alternatively, the authentication input device **1312** on the main entry system **1002** may be used for voice capture. The authentication input device **1312** may be equipped with a microphone that may be mounted near the barrier **1110**.

[0108] Each user may use the first user device **1006a** as the first factor for identity input. The first user device is assigned a unique identifier which is transmitted to the main entry system **1002** when the user uses the first user device **1006a** to request access. Alternatively, the user may be assigned a code to be entered into the main entry system **1002** to request access. The assigned code may be

entered, e.g., on a keypad provided by the identity input device **1310** on the main entry system **1002**. The unique identifier is the user's unique value that is used for identity input verification. Depending on how the user's input methods are configured, it is either the unique identifier provided on the first user device **1006a** or the manual code that the user enters.

[0109] The administration device **1104** is used by administrators for configuring and monitoring the main entry system **1002** and/or the authorization system **1004**. The administration device **1104** may be a computer, smart phone, or tablet connected wirelessly or by wire to the network **1100** via a network interface **1500**, which is similar to the previously described network interfaces **1300**, **1400**. The administrative device **1104** also preferably includes one or more of the following modules: a user interface **1502**, a database **1504**, and a notification handler **1506**. The user interface **1502** is used by the administrator to access and configure the main entry system **1002** or the authorization system **1004**. The database **1504** is used to store system configurations and load user data. The notification handler **1506** is used to receive notifications of abnormal events observed by the authorization system. An example of an event would be when a user has failed the authorization checks multiple times.

[0110] The system **1000** may also include a remote questionnaire pointer (RQP) **1102** which is preferably a permanently fixed device, such as a sign, that may be connected to the network **1100** (hence the broken line connecting the RQP **1102** with the network **1100**). Examples of the QP **1102** may be a parking sign, a sign along a walkway, or a sign at the barrier **1110** that provides instructions to the user seeking to enter a building/parking lot. The RQP **1102** preferably provides a URL link to a questionnaire/survey that visitors of a building must fill out before entry. The sign may be an electronic sign that is connected to the network **1100**. The QP **1102** provides the URL link by giving a static QR Code or phone number for receiving the URL via text. More advanced implementations may broadcast the URL via a radio embedded into the device. In the latter, the sign may be equipped with a solar panel to power the radio. The RQP **1102** may be controlled by the access controller similarly to the one QP **1314** described above for questionnaire delivery.

[0111] FIG. **12** is a flow chart showing a high-level operation of the exemplary entry system **1000**. Once the system is configured by the administrator (box **1600**), the system **1000** may be operated to control access to a secure space **114**. Once the user presents his/her unique identifier (identity input) to the main entry system **1002** (box **1602**), e.g., by scanning his/her first user device **1006a** or entering his/her code into the main entry system **1002**, the main entry system **1002** sends the unique identifier to the authorization system **1004** (box **1604**) for verification by the questionnaire application **1410** (box **1606**). If the identity check fails, entry is denied (box **1610**). If the identity check passes, the questionnaire is sent to the user if one is required (box **1608**). Once the user responds to the questionnaire (box **1612**), the response is sent to the authorization system **1004** for authorization. If the response fails, access is denied. If the response passes, the authorization system **1004** determines whether additional authentication is needed (box **1614**). If not, access is granted (box **1616**). If additional authorization is needed, the user is presented with additional authentication input (box **1618**), e.g., facial and/or voice recognition. Once the user enters the additional authentication input, it is verified by the authorization system **1004** (box **1620**). If verification fails, access is denied (box **1610**); if verification passes, access is granted (box **1616**).

[0112] FIGS. **13** and **14** shows the questionnaire configuration and the authorization configuration, respectively. Referring to FIG. **13**, the administrator uses the administrator device **1104** to connect to the authorization system **1004** and the main entry system **1002** via the network **1100** (box **1700**). Once connected to the authorization system **1004**, the administrator uses the questionnaire application **1410** to set the questions to be presented to the user (box **1702**). For each question, the administrator configures it as an authorization check or not (box **1704**). The administrator then sets the required answer(s) for each question that is configured as an authorization check (box **1706**). The administrator then configures the methods for delivering the questionnaire to the user (box **1708**), e.g., by geofencing, SMS, passive pointer, and/or broadcast pointer, and the questionnaire

expiration date (box **1710**). The administrator then configures the QP **1314** used to display the questionnaire to the user (box **1712**). The configuration process may simply a drop down menu with a list of RQPs and main entry systems for the administrator to select and configure. The questionnaire application **1410** then stores the questionnaire and associated configuration data in the database **1402** on the authorization system **1004** (box **1714**). The process is then repeated until all required questionnaires are configured.

[0113] Referring to FIG. **14**, the administrator uses the administrator device **1104** to connect to the authorization system **1004** via the network **1100** (box **1750**). The administrator then uses the identity and authentication application **1406** on the authorization system **1004** and the database **1504** on the administrator device **1104** to upload each user's unique identifier, name, and raw authentication input to the database **1402** of the authentication system **1004** (box **1752**). The administrator then uses the identity and authentication application **1406** to configure the input methods for entry system **1002** used to captures the user's identity and authentication input (box **1754**). The identity and authentication application **1406** then uses the generator **1408** to securely store the authentication input as a hash value (box **1756**). The administrator then uses the authorization application **1412** to set the authorization checks required for each particular user or user group to gain access (box **1758**). The authorization application **1412** next stores a user data object for each user in the database **1402** of the authentication system **1004** (box **1760**) which includes the user's unique identifier, name, authentication input hash, input methods, and associated authorization checks. The authorization application **1412** may then upload the database of user data objects to the main entry system **1002** via the network **1100**. Uploading to the user data objects to the main entry system **1002**, however, is an alternative approach. Ordinarily, the user data objects are stored only in the authentication system **1004**.

[0114] FIGS. **15-18** show methods for the exemplary system **1000** to deliver the questionnaire to the user, including via geofencing, passive pointer, radio broadcast pointer, and SMS, respectively. Referring to FIG. **15**, which is a flow chart showing a method for geofencing delivery of the questionnaire, the administrator enables the geofencing delivery method (box **1800**) and the geofencing perimeter (box **1802**) using the questionnaire application **1410**. To use geofencing delivery, the user must enable the questionnaire viewer **1216**, the location service **1206**, and notifications **1210** on the second user device **1006b** (box **1804**). The user must also login to the questionnaire viewer **1216** (box **1806**) and have the login information validated by the questionnaire application **1410** in the authorization system **1004** (box **1808**). Once logged in, the questionnaire viewer **1216** uses the network interface **1200** on the second user device **1006b** to retrieve the questionnaire from the questionnaire application **1410** (box **1812**). The questionnaire viewer **1216** configures the OS location service **1206** on the second user device **1006b** to provide a push notification when the second user device **1006b** is within the geofencing perimeter (box **1814**). This configuration is preferably accomplished at the time the questionnaire viewer **1216** is installed on the second user device. Once within the geofencing perimeter, the questionnaire viewer **1216** informs the user that he/she is within the perimeter and checks to determine whether the user needs to fill out a questionnaire by checking the expiration date and completion status of the questionnaire (box **1818**). If there is no expired or incomplete questionnaire, no action is taken (box **1816**). Otherwise, the questionnaire viewer **1216** uses the OS notification handler to notify the user that a questionnaire is required to be successfully completed (box **1820**). Once, the user presses the notification (box **1822**), the questionnaire viewer **1216** presents the user with the required questionnaire(s) (box **1824**). Once the user completely responds to the questionnaire, the questionnaire viewer **1216** sends the completed response to the questionnaire application **1410** to be processed and stored (box **1826**).

[0115] Referring to FIGS. **16a** and **16b**, which show a method for delivery of the questionnaire by passive pointer, the administrator, using the questionnaire application **1410**, enables the passive delivery method (box **1830**) and configures the particular main entry system **1002** and RQP **1102** to

be associated with the questionnaire (box **1832**). Once enabled and configured, the main entry system **1002** and the RQP **1102** enables digital displays of the passive pointer (box **1834**). The authorization system **1004** then provides the main entry system **1002** and the RQP **1102** with content to display on the digital display (box **1836**). The main entry system **1002** and the RQP **1102** display the passive questionnaire pointer on their digital display (box **1838**). The user then reads the content shown on the digital display of the main entry system **1002** or the RQP **1102** informing him/her a questionnaire is required (box **1840**). Using the second user device **1006b**, the user can choose to complete the questionnaire (box **1842**) by manually entering the URL or by scanning the QR code on the digital display. To manually enter the URL, the user may use the browser **1212** on the second user device **1006b** to enter the URL presented on the main entry system **1002** or the RQP **1102** (box **1844**). To scan the QR code, the user uses the camera **1208** on the second user device **1006b** to scan the QR code presented by the main entry system **1002** or the RQP **1102** (box **1846**) which provides the URL present in the QR code to the browser **1212** on the second user device **1006b** (box **1848**). In either case, the browser **1212** opens the URL which is lead to the questionnaire application **1410** running on the authorization system **1004** (box **1850**). Using the data encoded in the URL, the questionnaire application **1410** presents the user with the require questionnaire(s) on the browser **1212** on the second user device **1006b** (box **1852**). Once the user responds to the questionnaire(s), the questionnaire application **1410** processes and stores the results in the database **1402** (box **1854**).

[0116] Referring to FIG. **17**, which is a flow chart showing a method for delivery of the questionnaire by broadcast pointers, the administrator, using the questionnaire application **1410**, enables the broadcast delivery method (box **1860**) and configures the particular main entry system **1002** and RQP **1102** to be associated with the questionnaire (box **1862**). This method uses radio broadcast of the URL as discussed above. Once enabled and configured, the main entry system **1002** and the RQP **1102** enable radio broadcasting of the questionnaire pointer (box **1864**). To use broadcast delivery, the user must enable the questionnaire viewer **1216**, the location service **1206**, and notifications **1210** on the second user device **1006b** (box **1866**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. The user must also login to the questionnaire viewer **1216** (box **1868**) and have the login information validated by the questionnaire application **1410** in the authorization system **1004** (box **1870**). Once logged in, the questionnaire viewer **1216** configures the OS location service **1206** on the second user device **1006b** to periodically scan for the radio broadcast from the main entry system **1002** or the RQP **1102** (box **1872**). When the broadcast is detected, the questionnaire viewer **1216** uses the network interface **1200** on the second user device **1006b** to provide the questionnaire application **1410** on the authorization system **1004** with the unique identifier of the questionnaire pointer and user information (box **1874**). If there is no expired or incomplete questionnaire, no action is taken (box **1875**). Otherwise, the questionnaire application **1410** sends the required questionnaire(s) to the question questionnaire viewer **1216** running on the second user device **1006b** (box **1876**). Once the questionnaire(s) is received, the questionnaire viewer **1216** uses the OS notification handler **1204** to notify the user (box **1878**). Once, the user presses the notification (box **1880**), the questionnaire viewer **1216** presents the user with the required questionnaire(s) (box **1882**). Once the user completely responds to the questionnaire, the questionnaire viewer **1216** sends the completed response to the questionnaire application **1410** to be processed and stored (box **1884**).

[0117] Referring to FIGS. **18a** and **18b**, which are a flow charts showing a method for delivery of the questionnaire by text messaging (a.k.a. SMS). The administrator, using the questionnaire application **1410**, enables the SMS delivery method (box **2000**) and configures the particular main entry system **1002** and RQP **1102** to be associated with the questionnaire (box **2002**). Once enabled and configured, the QP **1314** of the main entry system **1002** and/or the RQP **1102** enable their

digital displays for the SMS request (box **2004**). The authorization system **1004** then provides the RQP **1102** or QP **1314** with content to display on the digital display (box **2006**). The RQP **1102** or QP **1314** then display the SMS request on their digital displays (box **2008**). The user then reads the content shown on the digital display of the QP of the or the RQP **1102** informing him/her that a questionnaire is required (box **2010**). Using the second user device **1006b**, the user can choose to complete the questionnaire (box **2012**) by requesting the questionnaire by following the instructions on the QP **1314** or the RQP **1102** (box **2012**) and texting the location ID of the RQP or QP to the number specified by the instructions (box **2014**). The messaging service **1404** of the authorization system **1004** receives the text message and parses its contents for the location ID of RQP or QP and sends the location ID to the questionnaire application **1410** (box **2016**). The questionnaire application **1410** determines, by looking to the database **1402**, whether that location ID is valid and whether any questionnaire(s) is associated with that location (box **2017**). If the user provided location ID matches and one or more questionnaires are required, then the location ID is valid, and the messaging service **1404** replies back to the text message with an URL to access the questionnaire and instructions for filling out the questionnaire (boxes **2018** and **2020**). The user can press on the URL link in the text message or copy and paste the URL in the browser **1212** on the second user device **1006b** (box **2021**) to access the questionnaire. The browser **1212** then opens the URL which is lead to the questionnaire application **1410** running on the authorization system **1004** (box **2022**). Using the data encoded in the URL, the questionnaire application **1410** presents the user with the require questionnaire(s) on the browser **1212** of the second user device **1006b** (box **2024**). Once the user responds to the questionnaire(s), the questionnaire application **1410** processes and stores the results in the database **1402** (box **2026**).

[0118] FIG. **19** is a flow chart showing the identity input process the user must follow to gain access to the secure space **114**. The identity input process is generally discussed above and shown in box **1602** of FIG. **12**. The administrator uses the authorization application **1412** (accessed via the administrator device **1104**) to select the kind of identity input to be used by the user (box **2100**). Two kinds of identity inputs may be selected: manual code entry (box **2116**) or first user device (box **2102**). The administrator selects one or both input methods for an individual user. The administrator may also select no identity for a particular user, in which case, no identity input is needed when that user requests entry to the secure space **114** (other requirements, such as questionnaire and/or facial or voice recognition may still be applicable to that user to gain entry).

[0119] For manual code input, the authorization application **1412** requests a new manual entry code for the user from the identity and authentication application **1406** (box **2118**), causing the identity and authentication application **1406** to generate a unique manual entry code (unique identifier) and to provide it to the authorization application **1412** (box **2120**). The authorization application **1412** stores the input method and entry code (unique identifier) in the user data object for that particular user (box **2122**), preferably in the database **1402** of the authorization system **1004**. The authorization application **1412** then displays the manual entry code to the administrator (box **2124**) who then provides it to the user (box **2126**). The user then manually enters the code into the identity input device **1310** of main entry system **1002** (box **2126**).

[0120] For using the first user device **1006a**, the authorization application **1412** requests a new first user device **1006a** unique identifier for the user from the identity and authentication application **1406** (box **2104**), causing the identity and authentication application **1406** to obtain the serial number and unique identifier of the first user device **1006a** from the database **1402** and to provide them to the authorization application **1412** (box **2106**). The authorization application **1412** stores the input method and unique identifier in the user data object for that particular user (box **2108**). The authorization application **1412** then displays the serial number of the first user device **1006a** to the administrator (box **2110**) who then provides the first user device to the user for use when requesting entry to the secure space **114** (box **2112**). To initiate entry request, the user scans the first user device **1006a** at the identity input device **1310** of main entry system **1002** (box **2114**).

[0121] In either event (manual code entry or use of the first user device **1006a**), the identity input device **1310** sends the received identity input to the access controller **1306** on the main entry system (box **2130**). The access controller **1306** then forwards the identity input to the authorization application **1412** running on the authorization system **1004** to initiate authentication check (box **2132**).

[0122] Once the identity input is completed, the system **1000**, if configured to do so for the particular user, conducts an authentication check to verify that the user who presented the identity input is allowed to gain entry to the secure space **114** (authentication). FIGS. **20-23** show different authentication checks that may be selected by the administrator for a particular user. The administrator may select approval from 1) the second user device **1006b** (FIG. **20**); 2) alphanumeric input (FIG. **21**); 3) proximity to the main entry system (FIGS. **22**); and 4) biometric data (FIG. **23**).

[0123] Referring to FIG. **20**, which is a flow chart showing the authentication input using the second user device **1006b** (box **2200**), the access controller **1306** disables the authentication input device **1312** on the main entry system **1002** (box **2202**). To initiate authentication, the user logs into the questionnaire viewer **1216** and enables notifications on the second user device **1006b** (box **2104**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. After receiving the identity input from the access controller **1306** (box **2132** in FIG. **19**), the authorization system **1004** sends a notification to the second user device **1006b** (box **2206**). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box **2208**). When the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instruction to provide approval for use of identity at the particular main entry system **1002** (box **2010**). The user either approves (box **2014**) the use or not (box **2012**) which is sent to the authorization application **1412** on the authorization system **1004**. In either case, the authorization application **1412** uses identity input from the access controller **1306** and authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box **2016**).

[0124] Referring to FIG. **21**, which is a flow chart showing the authentication input using alphanumeric value (box **2300**), the access controller **1306** disables the authentication input device **1312** (box **2302**), and enables the digital display of the QP **1314** on the main entry system **1002** (box **2304**). The user is shown an alpha numeric value on the digital display of the QP **1314** (box **2304**). The user then logs into the questionnaire viewer **1216** and enables notifications on the second user device **1006b** (box **2306**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the authorization system **1004** sends an alphanumeric code to the access controller **1306** on the main entry system **1002** to be displayed to the user on the digital display of the QP **1314** for the user (box **2308**), and sends a notification to the second user device **1006b** (box **2310**). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box **2312**). When the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instruction to enter the alphanumeric value provided on the digital display of the QP **1314** (box **2314**). The user then enters the alphanumeric value using the questionnaire viewer **1216** on the second user device **1006b** (box **2316**). The questionnaire viewer **1216** then sends the alphanumeric value entered by the user to the authorization application **1412** on the authorization system **1004** (box **2318**) which uses the identity input from the access controller and the alpha numeric value

input to determine whether access should be granted (box 2320).

[0125] Referring to FIG. 22, which is a flow chart showing the authentication check using the proximity of the second user device **1006b** to the main entry system **1002** (box 2300), the access controller **1306** disables the authentication input device **1312** (box 2302), and enables the radio on the QP **1314** of the main entry system **1002** for broadcasting (box 2304). The user then logs into the questionnaire viewer **1216** and enables notifications and location service on the second user device **1006b** (box 2406). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box 2132 in FIG. 19), the questionnaire viewer **1216** configures the OS location service **1206** of the second user device **1006b** to periodically scan for broadcasts from the QP **1314** (box 2408). Once a broadcast is detected, the questionnaire viewer **1216** uses the network interface **1200** of the second user device **1006b** to provide the authorization application **1412** on the authorization system **1004** with the unique identifier broadcasted by the QP **1314** (box 2410). That unique identifier is used as the authentication input. The authorization application **1412** then uses the identity input from the access controller **1306** and the authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box 2312).

[0126] Referring to FIG. 23, which is a flow chart showing the authentication check using biometric data (box 2300), the access controller **1306** disables the authentication input device **1312** (box 2302), and enables the digital display of the QP **1314** on the main entry system **1002** (box 2304). To obtain biometric input from the user, the administrator may use the authentication input device **1312** of main entry system **1002** or the second user device **1006b**. FIG. 24 is a flow chart showing the authentication check using biometric authentication input from the user device **1006b**; and FIG. 25 is a flow chart showing the authentication check using biometric authentication input from the authentication input device **1312** of the main entry system **1002**.

[0127] Referring to FIG. 24, if the administrator elects to use the second user device **1006b** to obtain biometric data, the access controller **1306** disables the authentication input device **1312** (box 2502). The user then logs into the questionnaire viewer **1216** and enables notifications and camera and/or microphone access on the second user device **1006b** (box 2504). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box 2132 in FIG. 19), the authorization system **1004** sends a notification to the second user device **1006b** (box 2506). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box 2508). Here, the administration can select facial recognition (box 2510) or voice recognition (box 2516) as the biometric data. If facial recognition is selected, when the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instructions to provide facial recognition using the camera **1208** on the second user device **1006b** (box 2512). The questionnaire viewer **1216** also enables the camera **1208** (box 2514) to take an image of the user's face. If voice recognition is selected by the administrator, when the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instructions to provide voice recognition using the microphone **1210** on the second user device **1006b** (box 2518). The questionnaire viewer **1216** also enables the microphone **1210** (box 2520) to record the speaking voice of the user. Once the image or voice is recorded, the questionnaire viewer **1216** uses a hash generator on the second user device **1006b** and the raw authentication input (image or voice recording) to generate an authentication hash value of the authentication input by sending the raw authentication input to the hash generator (box 2522). The hash generator on the second user device

1006b uses the raw authentication input to generate the authentication has value (box **2524**). The questionnaire viewer **1216** then sends the authorization hash value to the authorization application **1412** on the authorization system **1004** to be used as the authentication input (box **2526**). The authorization application **1412** then uses the identity input from the access controller **1306** and the hashed authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box **2528**).

[0128] Referring to FIG. 25, if the administrator elects to use the authentication input device **1312** of the main entry system **1002** to obtain biometric data, the access controller **1306** enables the authentication input device **1312** (box **2530**). Here, the administration can select facial recognition (box **2532**) or voice recognition (box **2536**) as the biometric data. If facial recognition is selected, after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. 19), the access controller **1306** enables the camera on the authentication input device **1312** on the main entry system **1002** (box **2534**) to take an image of the user's face. If voice recognition is selected, after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. 19), the access controller **1306** enables the microphone on the authentication input device **1312** on the main entry system **1002** (box **2538**) to record the talking voice of the user. Once the image or voice is recorded, the authentication input device **1312** send the raw authentication input (image or voice recording) to the access controller **1306** (box **2540**) which, in turn, sends the raw authentication input to the hash generator **1308** to generate a hash value of the raw authentication input (box **2542**). The access controller **1306** then sends the hash value to the authorization application **1412** on the authorization system **1004** to be used as the authentication input (box **2544**). The authorization application **1412** then uses the identity input from the access controller **1306** and the hashed authentication input to determine whether access should be granted.

[0129] FIG. 26 is a flow diagram showing the authorization check process after identity and authentication inputs have been obtained by the authorization application **1412** on the authorization system **1004**. Once the identity input is received, it is checked against the user data object for the particular user in the database **1402** (box **2600**). If the identity input does not match, access is denied. The administrator then uses the authorization application **1412** to select the types of authorization check required (box **2602**). If authentication is enabled, the authorization application **1412** receives the authentication input from the main entry system **1002** or the questionnaire viewer **1216** of the second user device **1006b** (see boxes **2528** of FIG. 24, **2412** of FIG. 22, **2320** of FIGS. 21, and **2216** of FIG. 20) and compares it to the user data object for the particular user in the database **1402** (box **2604**). If the authentication input does not match access is denied. The administrator then uses the authorization application **1412** to select the types of authorization check required (box **2606**). If questionnaire authorization check is enabled, the authorization system **1004** checks whether the authorization application **1412** receives valid response(s) to the questionnaire (box **2608**). If the response(s) is not valid, access is denied. If the response(s) is valid, the authorization application **1412** sends an access granted message to the access controller **1306** on the main entry system **1002** (box **2610**). The access controller **1306**, in turn, release the lock on the barrier to allow the user to enter the secure space **114** (box **2612**).

[0130] Although certain presently preferred embodiments of the invention have been specifically described herein, it will be apparent to those skilled in the art to which the invention pertains that variations and modifications of the various embodiments shown and described herein may be made without departing from the spirit and scope of the invention. Accordingly, it is intended that the invention be limited only to the extent required by the appended claims and the applicable rules of law.

Claims

- 1.** A method for controlling access to a secure space, comprising: receiving, at a main entry system, an identity input associated with a user; receiving, at the main entry system or a verification system, an authentication input from a user device associated with the user, wherein the authentication input comprises at least one of a facial image or a voice recording; comparing the authentication input to a stored authentication reference in a database associated with the verification system; verifying, by the verification system, that the user has completed a questionnaire previously sent to the user device; retrieving a user data object associated with the user, the user data object comprising access control criteria specifying conditions for access based on user-specific attributes; and determining, based on the identity input, a result of the comparing, a completion status of the questionnaire, and the access control criteria, whether to grant the user access to the secure space.
- 2.** The method of claim 1, wherein the authentication input is received in response to an authentication request transmitted from the verification system to the user device.
- 3.** The method of claim 1, wherein the questionnaire includes at least one health-related question and the completion status indicates that the user has affirmatively responded to all required questions.
- 4.** The method of claim 1, wherein the stored authentication reference comprises a hashed facial image or hashed voice recording corresponding to the user.
- 5.** The method of claim 1, wherein the access control criteria include one or more temporal constraints specifying allowed access times for the user.
- 6.** The method of claim 1, further comprising, after determining to grant access, transmitting a control signal from the verification system to the main entry system to unlock a barrier controlling physical entry to the secure space.
- 7.** The method of claim 1, wherein the determining step includes evaluating whether the identity input and the authentication input are associated with a user account stored in the database.
- 8.** A system for controlling access to a secure space, comprising: a main entry system configured to receive an identity input associated with a user; a user device associated with the user, the user device configured to provide an authentication input comprising at least one of a facial image or a voice recording; a verification system in communication with the main entry system and the user device, the verification system comprising a database and configured to: compare the authentication input to a stored authentication reference in the database; verify that the user has completed a questionnaire previously sent to the user device; retrieve a user data object associated with the user, the user data object comprising access control criteria specifying conditions for access based on user-specific attributes; and determine, based on the identity input, the result of the comparison, the completion status of the questionnaire, and the access control criteria, whether to grant the user access to the secure space.
- 9.** The system of claim 8, wherein the verification system is further configured to transmit an authentication request to the user device prior to receiving the authentication input.
- 10.** The system of claim 8, wherein the stored authentication reference in the database comprises a hashed biometric sample corresponding to the user.
- 11.** The system of claim 8, wherein the questionnaire includes at least one health screening question, and the verification system is configured to determine the completion status based on required responses to the health screening question.
- 12.** The system of claim 8, wherein the access control criteria in the user data object include a time window specifying when the user is permitted to access the secure space.
- 13.** The system of claim 8, wherein the main entry system is further configured to unlock a physical barrier in response to a determination to grant access by the verification system.
- 14.** The system of claim 8, wherein the verification system is further configured to associate the identity input and the authentication input with a user account stored in the database.

- 15.** A user device configured to facilitate access control to a secure space, the user device comprising: a sensor configured to capture an authentication input associated with a user, the authentication input comprising at least one of a facial image or a voice recording; a communication interface configured to: receive, from a remote verification system, a questionnaire for completion by the user; transmit, to the remote verification system, a response to the questionnaire; and transmit the authentication input to the remote verification system; a memory storing executable instructions; and a processor configured to execute the instructions to cause the user device to: associate the authentication input with a user identity; generate a data element indicating completion of the questionnaire; and transmit, to the remote verification system, the user identity, the authentication input, and the data element indicating questionnaire completion for use in an access determination.
- 16.** The user device of claim 15, wherein the sensor comprises a camera and a microphone, and the authentication input comprises both a facial image and a voice recording.
- 17.** The user device of claim 15, wherein the processor is further configured to encrypt the authentication input before transmission to the verification system.
- 18.** The user device of claim 15, wherein the processor is further configured to generate a hash of the authentication input, and the communication interface is configured to transmit the hash to the remote verification system.
- 19.** The user device of claim 15, wherein the data element indicating completion of the questionnaire comprises a digital signature generated by the user device.
- 20.** The user device of claim 15, wherein the processor is further configured to store a timestamped log of the authentication input and questionnaire response for audit purposes.
-